

# Joanna X. Ju

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## Education

### University of California, San Diego

*Bachelor of Science in Computer Engineering*

Cumulative GPA: 3.70 / 4.00 (UC San Diego & Pasadena City College)

La Jolla, CA

Expected June 2028

**Relevant Coursework:** Computer Architecture, Digital Circuits, Digital Systems, Systems Programming, Advanced Data Structures, Circuits & Systems, Assembly Language

## Projects

### Custom 8-bit RISC Processor Design

*SystemVerilog, Assembly, C++*

- Designed and implemented a custom 8-bit RISC processor with a 9-bit instruction set, 8-register file, ALU, condition flags, branching, and separate instruction and data memories.
- Developed synthesizable SystemVerilog RTL for the datapath, control logic, register file, ALU, and memory interfaces.
- Built a two-pass C++ assembler to translate custom assembly programs into machine code and resolve labels and branch offsets.
- Verified individual modules and full-processor functionality through simulation and FPGA testing using ModelSim/Quarta and Quartus.

## Experience

### High Performance Computing & Embedded Systems Intern

Sep 2026 – Present

*NASA Armstrong Flight Research Center (AFRC)*

Edwards, CA

- Support development of real-time and high-performance embedded software in C/C++ for aerospace computing and flight-control systems.
- Develop and test software in Linux and RTOS environments for deterministic, time-sensitive applications.
- Contribute to software verification, system integration, and hardware-in-the-loop (HIL) testing in safety-critical environments.
- Work with embedded communication protocols and interfaces including UART, CAN, Ethernet, SPI, and I2C.

### Software/Avionics Team Member

Oct 2025 – Present

*UC San Diego Rocket Propulsion Laboratory (RPL)*

La Jolla, CA

- Collaborate with the avionics team to maintain and validate embedded systems for student-built rockets.
- Analyze embedded firmware and hardware interfaces for flight-computer, ground-station, and sensor systems.
- Test system functionality and diagnose hardware-software integration and communication issues.

### SPIS Mentor

Aug 2026 – Sep 2026

*UC San Diego, Computer Science and Engineering Department*

La Jolla, CA

- Mentored incoming computer science and engineering students in algorithmic problem-solving, programming, and debugging.
- Guided student teams in designing, programming, and debugging Raspberry Pi robotics projects using Python.
- Supported introductory computing coursework and provided technical feedback to students throughout the program.

### Computer Science Tutor

Aug 2024 – Jun 2025

*Pasadena City College*

Pasadena, CA

- Tutored students in computer science coursework covering programming, data structures, algorithms, and debugging.
- Explained programming concepts and debugging strategies through individualized and small-group academic support.

### Math Instructor

Jul 2024 – Aug 2026

*Mathnasium*

Temple City & San Diego, CA

- Instructed students using individualized mathematics curricula across a range of grade levels and mathematical backgrounds.
- Explained mathematical concepts and problem-solving strategies while adapting instruction to individual student needs and progress.

## Skills

**Languages:** C, C++, Python, Java, SystemVerilog, Verilog, Assembly

**Hardware:** RTL Design, FPGA, Computer Architecture, Processor Design, Digital Logic

**Embedded:** Microcontrollers, Arduino, Raspberry Pi, UART, CAN, Ethernet, SPI, I2C

**Tools:** Linux, Git, GitHub, Quartus, ModelSim/Quarta, Cadence Virtuoso, PSpice, LTspice, VS Code